

## AERODYNAMICS LAB

| III Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |              |   |   |         |               |     |       |
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| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6AE07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | PCC      | L            | T | P | C       | CIA           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          | -            | - | 3 | 1.5     | 40            | 60  | 100   |
| LIST OF EXPERIMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |              |   |   |         |               |     |       |
| <div>1. Calibration of Venturi meter</div> <div>2. Determination of Coefficient of discharge for a small orifice by a constant head method.</div> <div>3. Determination of Coefficient of loss of head in a sudden contraction and friction factor.</div> <div>4.Verification of Bernoulli's equation</div> <div>5. Calibration of subsonic wind tunnel.</div> <div>6 Pressure distributions over smooth and rough cylinder.</div> <div>7. Pressure distribution over cambered airfoils</div> <div>8 Flow visualization studies over sphere.</div> <div>9. Flow visualization studies over cylinders.</div> <div>10 Flow visualization studies in low-speed flows over airfoil with different angle of incidence.</div> <div>11. Drags estimation on airfoils</div> <div>12. Drags estimation on blunt bodies</div> <div>Note: Ten experiments should be performed.</div> |          |              |   |   |         |               |     |       |
| Reference Books:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |              |   |   |         |               |     |       |
| <div>1. John. M. Cimbala Yunus A. Cengel (2019), Fluid Mechanics: Fundamentals and Applications, 4<sup>th</sup> edition, McGraw-Hill, ISBN: 9353166217</div> <div>2. Jewel B. Barlow(1999), Low-Speed Wind Tunnel Testing, 3<sup>rd</sup> edition, Wiley-Interscience, ISBN: 0471557749</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |              |   |   |         |               |     |       |
| COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |              |   |   |         |               |     |       |
| <div>1. Demonstrate a comprehensive understanding of fluid flow measurement techniques.</div> <div>2. Determine coefficients, losses, and pressure distributions in various fluid flow scenarios.</div> <div>3. Analyze and understand fluid flow behavior, pressure changes, and velocity profiles in different situations.</div> <div>4. Analyze airflow patterns and pressure distributions over various shapes and angles.</div> <div>5. Assess how the shape and design of objects affect their resistance to fluid flow.</div>                                                                                                                                                                                                                                                                                                                                      |          |              |   |   |         |               |     |       |